



Engineering Project Management

Team – 2 EnviroMentalists

PIEZOELECTRIC WALKWAYS AT NORTHEASTERN UNIVERSITY

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Problem Statement

Northeastern University's high-traffic path between Krentzman Quad and Snell Library wastes significant kinetic energy daily. In the face of a global climate crisis, this project aims to harness that energy using piezoelectric technology, providing a sustainable power source for campus lighting while reducing reliance on fossil fuels.



Solution

The project proposes installing a piezoelectric walkway using Pavegen tiles along the path from Krentzman Quad to Snell Library. This system will harness energy from footsteps and convert it into electricity to power the walkway's LEDs, integrating renewable energy into campus infrastructure. The solution also involves energy storage in lithium-ion batteries, enabling consistent lighting and reducing main grid dependency.

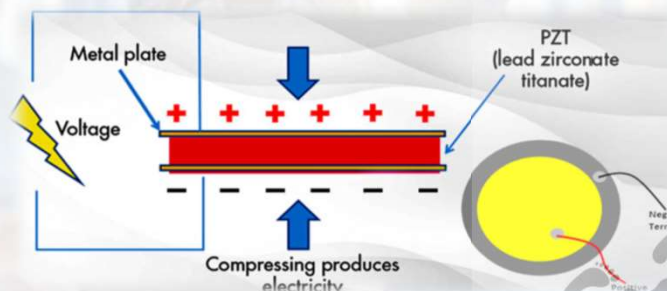


Goals and objectives

- Generate 5 watts of energy per footstep and more than 60 kWh of energy daily
- Reduce the carbon footprint by more than 2.9 tons per year
- Reduce electricity consumption and achieve cost savings
- Promote sustainability initiatives and student involvement



Technical Overview



Working of Pavegen tiles



Pavegen tiles

Piezoelectric materials generate an electric charge when subjected to mechanical stress, like footsteps on a walkway. This process occurs when pressure shifts the internal charged particles, creating an electric field. The energy harvested from these movements can be used to power nearby systems, reducing reliance on the electrical grid.

"Embrace the Future, It's Now"

Pavegen V3 tiles, designed for high-traffic areas, convert footsteps into electrical energy. Each tile measures 50 cm per side and generates 5 watts per step. The modular, durable composite material includes an anti-slip coating and electromagnetic generators. Wireless sensors track foot traffic and energy output, providing real-time data.

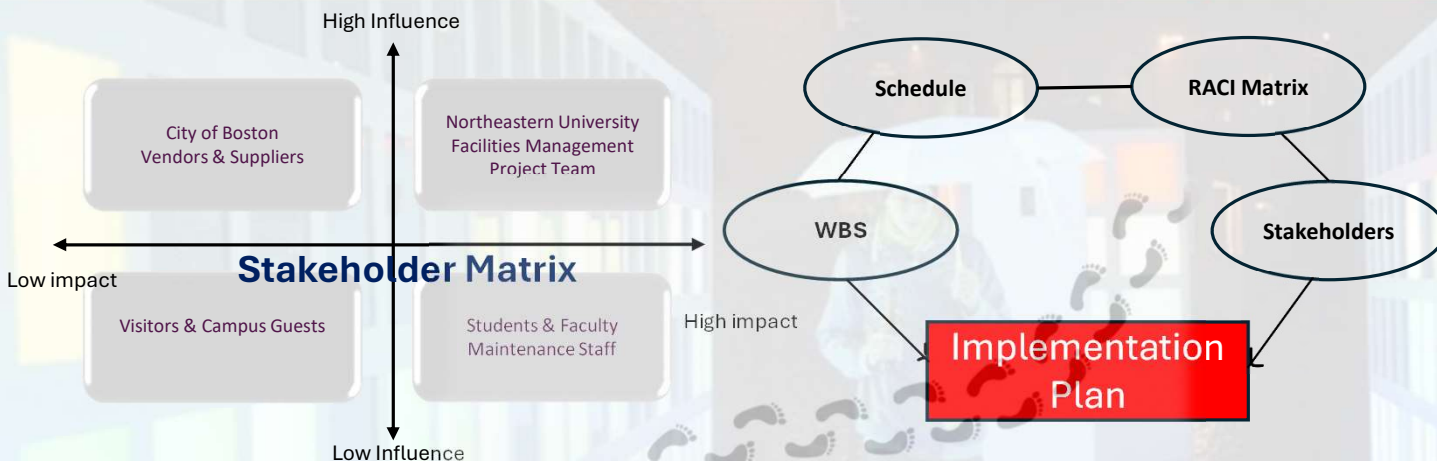


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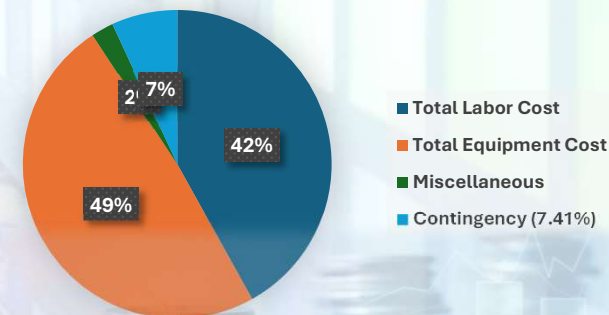
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BUDGET SUMMARY	
Resources	Total Cost
Total Labor Cost	\$715,444.20
Total Equipment Cost	\$833,432.00
a. Pavegen Tiles	\$243,432.00
b. Wiring and Conduits	\$5,250.00
c. Electrical Components	\$44,750.00
d. Battery Storage	\$540,000.00
Miscellaneous	\$40,000.00
Contingency (7.41%)	\$117,735.73
Total Cost (Pre-Rebate)	\$1,588,876.20
Rebate (Inflation Reduction Act, 20%)	(\$317,775.24)
Final Project Cost	\$1,388,836.69

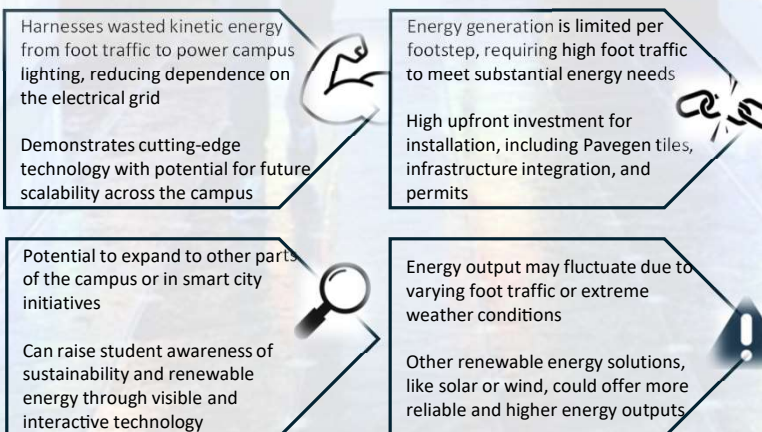
Budget Summary Chart



Start date
01/06/2025

The piezoelectric walkway project at Northeastern University will be completed over 3 months, with tasks broken into four distinct phases, and Gantt charts are used to track progress, identify delays, and ensure timely completion

End date
04/23/2025



SWOT Analysis